EDUCATION - NEWS CONSULT

7TH MOCK EXAMINATION MARKING SCHEME

COMPUTING

OBJECTIVES

1. C	**11.B**	**21.C**	**31.C**
2. B	**12.A**	**22.B**	**32.B**
3. C	**13.B**	**23.C**	**33.B**
4. B	**14.A**	**24.A**	**34.C**
5. C	**15.B**	**25.A**	**35.A**
6. C	**16.C**	**26.C**	**36.A**
7. C	**17.A**	**27.C**	**37.D**
8. A	**18.D**	**28.B**	**38.B**
9. C	**19.B**	**29.C**	**39.A**
10.B	**20.B**	**30.A**	**40.B**

QUESTION 1

a. i. Identify the topologies labelled A, B, C and D

A - Hybrid topology

B - Tree topology

C - Ring topology

D - Dual ring topology ***1 mark each × 4 = 4 marks***

ii. State one feature each of the diagrams labelled above.

A - Hybrid Topology

* Combination: Utilizes a mixture of two or more different network topologies.
* Scalability: Offers flexibility for expansion and adaptation to varying network requirements.
* Reliability: Provides redundancy and fault tolerance by incorporating multiple topology structures.
* Complexity: Can be more complex to design, implement, and manage compared to single-topology networks.

B - Tree Topology:

* Hierarchy: Organized in a hierarchical structure with a root node at the top and branches extending downward.
* Centralization: Features a central node (root) that controls the flow of communication.
* Scalability: Supports expansion by adding more branches or nodes to the existing structure.
* Dependency: Relies heavily on the central node, making it vulnerable to single point of failure.

C - Ring Topology:

* Ring Structure: Nodes are connected in a circular arrangement, where each node is connected to exactly two other nodes.
* Token Passing: Utilizes token passing mechanism for data transmission, where only the node with the token can send data.
* Simple Configuration: Relatively simple to configure and implement compared to other topologies.
* Failure Impact: Susceptible to network disruption if any node or connection in the ring fails, causing the entire network to be affected.

D - Dual Ring Topology:

* Redundancy: Features two interconnected rings, providing redundancy and fault tolerance.
* Data Transmission: Utilizes one ring for data transmission while the other serves as a backup.
* Increased Reliability: Offers increased reliability compared to a single ring topology due to redundancy.
* Complexity: Can be more complex to implement and manage compared to single-ring topologies due to dual infrastructure.

1 mark each × any 4 = 4 marks

b. **i. Provide the names of the cloud storage platforms below.** (3 marks)

I - Dropbox
II - iCloud
III - OneDrive ***1 mark each × 3 = 3 marks***

ii. State three threats that cause data corruption.

* Malware
* Hardware failure
* Software bugs
* Power outages
* Human error
* Improper shutdown
* Wrong ejection of storage devices ***1 mark each × any 3 = 3 marks***

iii. **Give three functions of the transport layer in the Open System Interconnection (OSI) model.**

* Segmentation and reassembly: Breaking data into smaller units for transmission and reassembling them at the destination.

* Error detection and correction: Identifying and correcting errors that may occur during data transmission.

* Flow control: Regulating the flow of data between sender and receiver to prevent congestion and ensure smooth transmission.

* Multiplexing: Managing multiple data streams over a single network connection by assigning unique identifiers to each stream.

* Connection establishment and termination: Setting up and tearing down communication sessions between sender and receiver.

2 mark each × 3 = 6 marks

iv. **Define information security.**

refers to the practice of protecting information and data from unauthorized access, use, disclosure, disruption, modification, or destruction.

2 marks

v. State two features of the Charms bar.

* Search
* Share
* Devices
* Settings
* Start menu 1 ***mark each × any 2 = 2 marks***

2. **a. What is digital footprint?**

A digital footprint refers to the trail of data and information that individuals leave behind as they interact with digital platforms and devices.

2 marks

b. Explain the following types of digital footprint.

i. Passive

A passive digital footprint refers to the data and information that is collected about an individual without their direct input or action.

This type of footprint is typically generated through automated processes and tracking mechanisms implemented by websites, apps, and online services.

2 marks

ii. **Active**

An active digital footprint refers to the data and information that individuals intentionally create and share through their online activities and interactions.

Unlike passive footprints, active footprints are generated through direct actions and inputs from the individual.

2 marks

c. i. List any four Artificial Intelligence softwares.

* TensorFlow
* PyTorch
* scikit-learn
* OpenAI GPT
* IBM Watson
* Microsoft Azure Cognitive Services
* Amazon SageMaker 1 ***mark each × any 4 = 4 marks***

ii. Write down the procedure to resize a picture in PowerPoint.

Click and drag the corner sizing handles until the picture is the desired size.

2 marks

3. **a. i. What is a programming language?**

A programming language is a formal language comprising a set of instructions used to produce various kinds of output. It enables a programmer to communicate with a computer by providing commands and algorithms to perform specific tasks, such as data manipulation, computation, or control of hardware devices.

2 marks

ii. **State three uses of programming language.**

* Developing software applications
* Creating websites and web applications
* Writing scripts for automation and system administration
* Designing mobile applications

* Implementing algorithms for data analysis and machine learning

1 ***mark each × any 3 = 3 marks***

b. **i. Define social media sites.**

Social media sites are online platforms that enable users to create profiles, share content, interact with others, and participate in virtual communities.

2 marks

ii. **Outline three benefits of video sharing sites.**

* Video sharing sites allow content creators to reach a worldwide audience, breaking geographical barriers and expanding their influence.

* Users can access a wide variety of content ranging from educational videos to entertainment, music, news, and more, catering to diverse interests and preferences.

* These platforms facilitate engagement through likes, comments, shares, and subscriptions, fostering interaction between creators and viewers, as well as among viewers themselves.

* Content creators can monetize their videos through advertising, sponsorship deals, subscription models, and donations, providing them with opportunities to generate revenue from their content.

* Video sharing sites empower individuals and communities to express themselves creatively, share their stories, and connect with like-minded individuals, fostering a sense of belonging and community.

1 ***mark each × any 3 = 3 marks***

c. **List two search engines.**

Google
Bing
Yahoo
Baidu
DuckDuckGo
Yandex
Ask.com
AOL Search
Excite
Dogpile 1 ***mark each × any 2 = 2 marks***

4. a. i. What is file extension?

A file extension is a suffix attached to the end of a filename, usually separated by a period. It indicates the format or type of the file and is used by the operating system and applications to determine how to handle the file.

2 marks

ii. Give three importance of file extensions.

* Identification: File extensions help users and software quickly identify the type of file and determine how to handle it.

* Compatibility: Different file extensions correspond to different file formats, ensuring compatibility with specific software applications and platforms.

* Organization: File extensions assist in organizing and categorizing files by type, making it easier for users to manage their digital assets.

* Interoperability: File extensions facilitate interoperability between different software programs by providing a standardized way to recognize and interpret file formats.

* Security: File extensions can indicate potentially harmful files, such as executable files (.exe) or script files (.bat), helping users and security software detect and mitigate potential risks.

1 ***mark each × any 3 = 3 marks***

b. i. State two advantages of AI.

Automation
Improved efficiency
Decision making
Personalization
Predictive analytics 1 ***mark each × any 2 = 2 marks***

ii. Outline three requirements of a broader work environment.

* Diversity and inclusion initiatives
* Open and transparent communication
* Flexible work arrangements
* Health and wellness programs
* Continuous learning and development opportunities
* Recognition and reward systems
* Collaborative team environment

* Accessible technology and resources
* Safe and ergonomic workspace provisions

1 ***mark each × any 3 = 3 marks***

c. **List any two types of printers.**

* Inkjet printer
* Laser printer
* Thermal printer
* Dot matrix printer 1 ***mark each × any 2 = 2 marks***

5. **. Write down four characteristics of the fifth generation of computers.**

* Artificial Intelligence Integration
* Quantum Computing Capabilities
* Advanced Parallel Processing
* Natural Language Processing
*Biometric Security Features

1 ***mark each × any 4 = 4 marks***

b. **i. Give three ways of reducing the heat of a printer.**

* Ensure proper ventilation around the printer.
* Clean the printer's internal components regularly.
* Reduce the print resolution or speed settings.
* Use high-quality paper to minimize friction and heat generation.
* Avoid printing in bulk or long print jobs continuously.

1 ***mark each × any 3 = 3 marks***

ii. **List three wearable display technologies.**

* Health monitoring devices
* Virtual Reality (VR) Headsets
* Smartwatches
* Heads-up Displays
* Smart Glasses with Micro-LED Displays

1 ***mark each × any 3 = 3 marks***

c. State the uses of the following MS Excel functions.

i. **MAX** - It is used to find the maximum value within a dataset. ***1 mark***

ii. **AVERAGE -** Calculates the arithmetic mean of a range of cells or a set of specified values. ***1 mark***